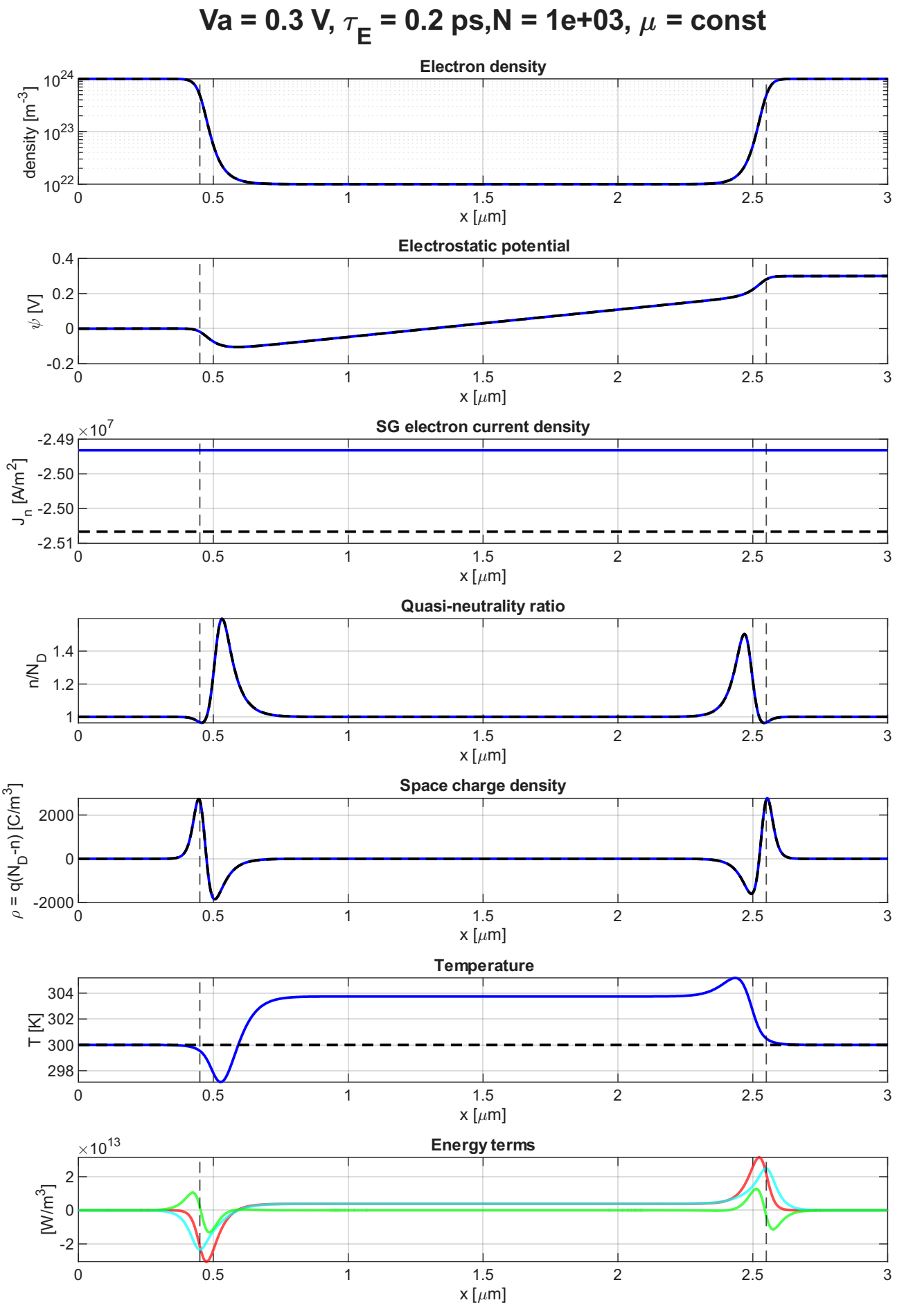

Va = 0.3 V, τ_E = 0.2 ps,N = 1e+03, μ = const
Electron density
density [m^{-3}]
10^{24}
10^{23}
10^{22}
0
0.5
1
1.5
2
2.5
3
x [μm]
Electrostatic potential
ψ [V]
0.4
0.2
0
-0.2
x [μm]
SG electron current density
$\times 10^7$
J_n [A/m^2]
-2.49
-2.50
-2.50
-2.51
x [μm]
Quasi-neutrality ratio
n/N$_D$
1.4
1.2
1
x [μm]
Space charge density
ρ = q(N$_D$-n) [C/m^3]
2000
0
-2000
x [μm]
Temperature
T [K]
304
302
300
298
x [μm]
Energy terms
$\times 10^{13}$
[W/m^3]
2
0
-2
x [μm]
Non-isothermal
Isothermal
$J_n \cdot E$
$\frac{3}{2} qn \frac{\theta - V_T}{\tau_E}$
$\frac{dS_n}{dx}$